**Chapter test**

# 40 Harmful effects of microorganisms

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

## A. Concept checking (10 marks, 1 mark each)

1. Which of the following correctly explain how microorganisms cause diseases in humans?

   (1) Some microorganisms produce toxins which have adverse effects on the host.
   (2) Some microorganisms produce antibodies to attack the host cells.
   (3) Some microorganisms produce enzymes to digest host tissues.

   A. (1) and (2) only
   B. (1) and (3) only
   C. (2) and (3) only
   D. (1), (2) and (3)

★ 2. Which of the following comparisons between food-borne infection and food poisoning is correct?

| | *Food-borne infection* | *Food poisoning* |
|---|---|---|
| A. | It usually has a shorter onset time. | It usually has a longer onset time. |
| B. | Botulism is an example of food-borne infection. | Salmonellosis is an example of food poisoning. |
| C. | It is caused by the growth of pathogens in contaminated food. | It is caused by microbial toxins in food. |
| D. | It cannot be prevented by thorough cooking. | It can be prevented by thorough cooking. |

★ 3. Which of the following statements about food poisoning is/are correct?

   (1) Salmonellosis is an example of food poisoning.
   (2) Ciguatera fish poisoning is caused by bacterial toxins.
   (3) The risk of ciguatera fish poisoning is higher when we eat large fish.

   A. (3) only
   B. (1) and (2) only
   C. (1) and (3) only
   D. (2) and (3) only


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4. Which of the following is an example of microbial deterioration?

A. biodegradation of plastic bags
B. production of blue cheese
C. rotting of fruits
D. decomposition of leaf litter ☐

★★5. Which of the following statements concerning the microorganisms that cause microbial deterioration is/are correct?

(1) They are decomposers.
(2) They are mainly fungi and viruses.
(3) All of them can cause diseases in humans.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only ☐

6. Which of the following is preserved by reducing its water content?

A. canned tuna
B. jam
C. kimchi
D. sausage ☐

★ 7. Which of the following about pasteurization is correct?

(1) Milk is heated to 72 $^{o}$C for 15 seconds.
(2) It does not kill all the bacteria in the milk.
(3) Refrigeration is not needed for pasteurized milk.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐


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★ 8. Which of the following microbial control methods make(s) use of heat to kill microorganisms?

(1) canning
(2) irradiation
(3) autoclaving

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only ☐

★★9. Which of the following foods will turn bad first at room temperature?

A. pickled cucumbers
B. frozen meat
C. biscuit
D. UHT milk ☐

★10. Which of the following statements about disinfectants is/are correct?

(1) They are more toxic than antiseptics.
(2) They cannot be used on living tissues.
(3) Iodine solution (5%) is a commonly used disinfectant.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only ☐

| Question | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Related section | 40.1 | 40.2 | 40.2 | 40.3 | 40.3 | 40.3 | 40.4 | 40.4 | 40.4 | 40.4 |


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## B. Problem solving (16 marks)

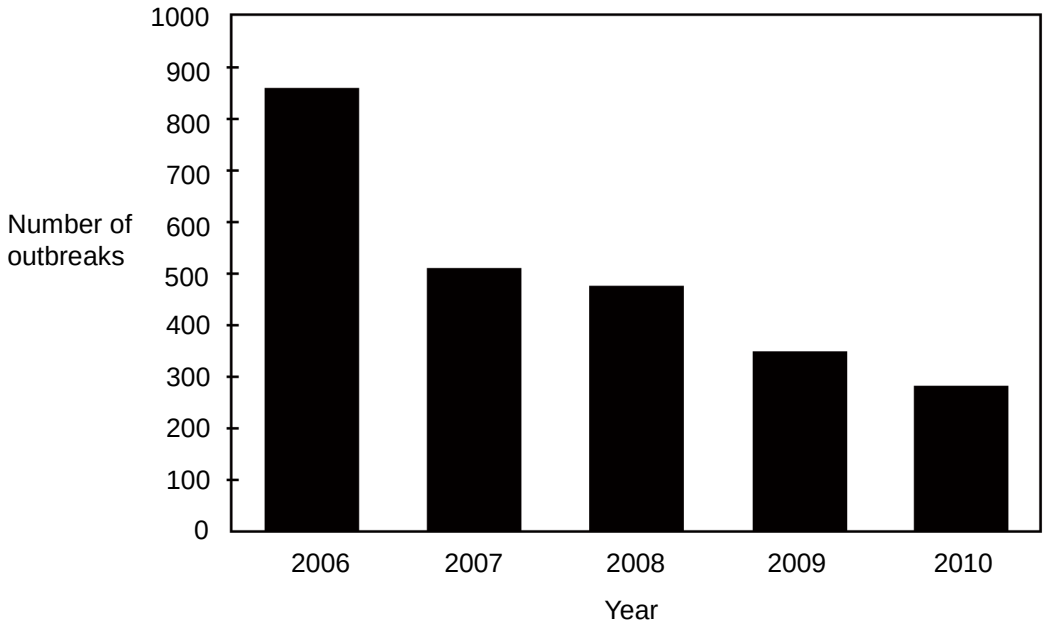
★ 11. Eggs are a high-risk food item for salmonellosis.

(a) Suggest ***one*** way in which the pathogen of salmonellosis can reach the food and causes the disease in humans. (3 marks)

(b) One effect of the toxins produced by the pathogen of salmonellosis is to cause epithelial cells lining the intestine to secrete sodium ions and chloride ions into the lumen. Suggest how this results in watery faeces. (2 marks)

(Total: 5 marks)

★ 12. The bar chart below shows the number of outbreaks of food poisoning in Hong Kong from 2006 to 2010.





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(a) What is food poisoning? (1 mark)

(b) Describe the trend in the number of outbreaks of food poisoning in Hong Kong from 2006 to 2010. Support your answers with data from the bar chart. (3 marks)

(c) Suggest ***two*** possible reasons to account for the trend you described in (b). (2 marks)

(d) It is expected that the number of food poisoning cases would rise in July and August each year. Suggest an explanation for this. (3 marks)

(e) Explain why thorough cooking is usually not so effective in preventing food poisoning. (2 marks)

(Total: 11 marks)


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## C. Exam challenge (14 marks)

★ 13. The foods below are preserved by different methods.

❶ 


pasteurized milk

❷ 


UHT milk

❸ 

canned fruits

❹ 

dried fruits

❺ 

frozen peas

(a) What is meant by microbial deterioration? (1 mark)

_____

(b) Suggest ***two*** substances other than food that are susceptible to microbial deterioration. (2 marks)

_____

_____


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(c) Milk shown in photographs ❶ and ❷ is preserved by two different methods.

(i) Describe how the milk was treated in these two methods respectively. (2 marks)

(ii) State ***one*** advantage and ***one*** disadvantage of these two methods respectively. (4 marks)

(d) What is the similarity between the preservation methods used in photographs ❸ and ❹? State the physical and biological process involved in these two methods respectively. (3 marks)

(e) The five foods shown are preserved by either killing microorganisms or inhibiting their growth. Classify these five foods into two categories. (2 marks)

By killing microorganisms:

By inhibiting the growth of microorganisms:

(Total: 14 marks)


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# Marking scheme

## A. Concept checking (10 marks, 1 mark each)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | B | 2. | C | 3. | A | 4. | C | 5. | A |
| 6. | B | 7. | A | 8. | C | 9. | B | 10. | B |

(1 mark each)

## B. Problem solving (16 marks)

11. (a) The egg shells are contaminated by the faeces of an infected chicken. The bacteria may enter the eggs when the eggs are being processed. (1)
When humans eat raw or inadequately cooked eggs contaminated with the bacteria, the bacteria enter the intestines of humans. (1)
The bacteria reproduce in their bodies and secrete toxins which cause the symptoms of salmonellosis. (1)
(Or other acceptable answers)

(b) Since the contents in the lumen have a lower water potential than the epithelial cells, (1)
water enters the contents in the lumen / leaves the epithelial cells by osmosis. (1)

(Total: 5 marks)

12. (a) Food poisoning is a disease caused by eating foods contaminated with toxins produced by microorganisms. (1)

(b) The number of outbreaks of food poisoning dropped significantly from 2006 to 2007. (1)
The number of outbreaks decreased gradually from 2007 to 2010. (1)
Quoting figures from the bar chart (e.g. the number of outbreaks dropped by about 43% from 2006 to 2007) (1)

(c) Any ***two*** of the following: (2)
- Increased awareness in food safety among the general public (1)
- Food safety was educated in schools (1)
- Improved food storage and food handling practices in the restaurants or food merchants (1)

(Or other acceptable answers)

(d) The average temperatures are higher in summer months, i.e. July and August, in Hong Kong. (1)
Microorganisms / Bacteria / Fungi grow faster at higher temperatures, (1)
leading to an increased production of microbial toxins and thus an increased chance of food poisoning. (1)


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(e) Food poisoning is caused by microbial toxins which are usually heat-stable. (1)
Even after thorough cooking, the toxins are still present in the food. / Heating cannot destroy the toxins in the contaminated food. (1)

(Total: 11 marks)

## C. Exam challenge (14 marks)

13. (a) Microbial deterioration is the decomposition of the materials that are useful to humans by microorganisms. (1)

(b) Any ***two*** of the following: (2)
- Cotton (1)
- Paper (1)
- Leather (1)
- Timber (1)

(Or other acceptable answers)

(c) (i) Photograph ❶: The milk is heated to 72 $^{o}$C for 15 seconds. (1)
Photograph ❷: The milk is heated to 138 $^{o}$C for two to four seconds. (1)

(ii) Photograph ❶ (pasteurization) (max. 2)

| Advantage | Disadvantage |
|---|---|
| • Taste or texture of milk is maintained. (1)<br>• Most nutrients in the milk are kept. (1)<br>(Or other acceptable answers) | • Shorter shelf life. (1)<br>• Refrigeration is needed. (1)<br>• The treatment does not kill all the microorganisms in the milk. (1)<br>(Or other acceptable answers) |

Photograph ❷ (ultra high temperature (UHT) treatment) (max. 2)

| Advantage | Disadvantage |
|---|---|
| • Longer shelf life. (1)<br>• It can be stored at room temperature if properly sealed. (1)<br>(Or other acceptable answers) | • Taste or texture of milk may be changed after the treatment. (1)<br>• A small amount of vitamins / nutrients may be lost during processing. (1)<br>• Heat treatment may change the structure of milk proteins, which may in turn affect the digestion of milk proteins in the human body. (1)<br>(Or other acceptable answers) |


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(d) Similarity: Water is removed from food. / By dehydration / By lowering water availability (1)

Process involved in preserving canned fruits: Osmosis (1)

Process involved in preserving dried fruits: Evaporation (1)

(e) By killing microorganisms: Pasteurized milk, UHT milk, canned fruits and dried fruits (1)

By inhibiting the growth of microorganisms: Frozen peas (1)

(Total: 14 marks)


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